

Response to reviewers — point by point

Trajectory Regularity Does Not Survive Decoding: Curvature and Few-Step Sampling in Latent Flow Matching

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BeNTo workshop @ NeurIPS 2026 — camera-ready preparation

Summary of disposition

Nine distinct weaknesses were raised across the two reviews. Both reviewers converged independently on three of them, which we treat as the binding items. One is now resolved by a new experiment; four are resolved by revision; four are conceded as genuine limitations and stated as such.

ID	Raised by	Weakness, in brief	Disposition
W1	R1, R2	Proposition 1 is a textbook bound, used qualitatively; “predicts” overstates it	Fully addressed
W2	R1, R2	Schedules test arc length, a first-order quantity, not the second-order term in the bound	Fully addressed (new experiment)
W3	R1, R2	One dataset, one autoencoder family; generality unknown	Remains a limitation
W4	R1	Intervals reflect sampler noise only; likely too tight	Remains a limitation
W5	R1	Compute checklist contradicts the appendix	Fully addressed
W6	R2	Reflow changes the velocity field and coupling, not only straightness	Partially addressed
W7	R2	MPJPE undefined in the main text	Fully addressed
W8	R2	Interventions support but do not isolate geometry	Partially addressed
W9	R1	Stratified analysis residual FID floor (<i>acknowledged as a strength; no action</i>)	—

The new experiment referred to under W2 is reported in full in the accompanying document *Curvature-matched step schedules* (job 405244, 29 September 2026; 512 clips $\times$ 5 seeds). Its headline numbers are quoted where relevant below.

1 Reviewer 1 (Rating 4, Confidence 3)

W1 — The theory is overstated

“Proposition 1 is a textbook bound. The fact that mapped curvature is absent from one upper bound does not mean it carries no signal, so ‘what Proposition 1 predicts’ should read ‘consistent with’.”

Response. We accept this without reservation. The reviewer’s logical point is correct: absence of a term from an upper bound does not establish that the corresponding quantity carries no information. Our phrasing claimed more than the analysis supports.

Exact change. Replace every instance of predictive phrasing about Proposition 1 with consistency phrasing. Specifically:

- §6, opening paragraph: “matching a standard Euler error bound in which the decoded curvature does not appear” $\rightarrow$ “consistent with a standard Euler error bound, in which the decoded curvature does not appear.”
- §6, paragraph beginning “Both quantities in the bound”: “The one absent from it does not” $\rightarrow$ “The one absent from it does not predict here, although its absence from an upper bound does not by itself imply that it carries no signal.”
- Abstract, sentence 4: same substitution of “consistent with” for “matching”.
- §3, after Proposition 1: add one sentence — “Proposition 1 is a standard global error bound for explicit Euler composed with a Lipschitz map; we use it to identify which quantities *can* enter the error, not to predict which do. The measurements in §6 decide that.”

Location. Abstract (1 sentence); §3 (1 new sentence after Prop. 1); §6 (2 sentences); §9 “The error analysis is an upper bound” paragraph already states this correctly and needs no change.

Status. Fully addressed — the change is purely one of claim strength, and the reviewer’s own wording is adopted.

W2 — The schedule test uses the wrong statistic

“The paper also argues that Euler error is second-order, yet it tests only first-order arc-length schedules. A curvature-based schedule is the informative missing experiment.”

Response. The reviewer is right, and the point identifies a genuine gap rather than a presentational one. For explicit Euler with step h_k , the local truncation error is $\text{LTE}_k \approx \frac{1}{2}h_k^2\|\ddot{Y}_{t_k}\|$, so equalising error across steps requires $h_k \propto \|\ddot{Y}\|^{-1/2}$. Our submitted schedule equalised arc length, giving $h_k \propto \|\dot{Y}\|^{-1}$: a different derivative and a different power. Arc length allocates steps by speed; the bound allocates them by difficulty.

Action taken. We ran the experiment. Error density $\rho(t) = \|\ddot{Y}_t\|^{1/2}$, with $\ddot{Y}$ estimated by second differences along the 50-step reference trajectory at no additional network cost, from 128 calibration clips drawn with noise independent of evaluation. Three arms: **A** curvature in $\mathcal{Z}$ (shared grid), **B** curvature in $\mathcal{Q}$ (shared grid), **C** curvature in $\mathcal{Q}$ calibrated per clip. Two controls: per-clip in $\mathcal{Z}$, and a same-noise oracle for C.

Outcome. The null was the statistic, not the idea.

Model	Paired difference	$n=2$	$n=4$	$n=8$	$n=16$
LFM	A – B	$-0.999 \pm .089$	$-0.978 \pm .068$	$-0.113 \pm .014$	$-0.018 \pm .004$
LFM	A – cosine	$+0.215 \pm .016$	$+0.074 \pm .013$	$-0.030 \pm .015$	$-0.007 \pm .006$
LFM	A – linear	$-1.847 \pm .182$	$-1.413 \pm .100$	$-0.445 \pm .034$	$-0.079 \pm .010$
CDFM	A – B	$-0.588 \pm .037$	$-0.269 \pm .043$	$-0.035 \pm .009$	$-0.007 \pm .007$
CDFM	A – cosine	$-0.214 \pm .014$	$-0.087 \pm .013$	$-0.032 \pm .012$	$-0.009 \pm .005$

Every A–B interval excludes zero except CDFM at $n = 16$, where discretisation error is already small. Curvature scheduling improves on linear by 1.85 and 1.41 FID on LFM at 2 and 4 steps, where the arc-matched grid stayed within a few percent of linear. Arm A also beats the cosine schedule from 8 steps on LFM and at every budget on CDFM.

Exact change. Rewrite §7 around the curvature result:

- New §7.1 “Allocating steps by the error term”: the derivation above, the estimator, and the go/no-go validation (rank correlation between predicted $\frac{1}{2}h^2\|\ddot{Y}\|$ and directly measured one-big-step error: 0.86–0.95 in $\mathcal{Z}$, 0.63–0.94 in $\mathcal{Q}$).
- Table 4 replaced by the new table with all schedules and both models; arms A, B, C and the two controls added as rows.
- New Figure: $\|\ddot{Y}_t\|$ against t , and the 8-step grids, showing why cosine is a strong baseline (it packs steps at the noise end, where curvature is largest) and how the $\mathcal{Z}$ -curvature grid additionally refines the data end.
- The arc-length grids demoted to a single paragraph, stated as the wrong statistic with this experiment as the evidence.
- §9: remove “a curvature-based construction in the decoded space might succeed where ours did not” — it has now been tested.

Location. §7 in full (rewritten); Table 4 (replaced); one new figure in §7; §9 limitations (one sentence removed); Appendix (estimator definition and validation table).

Status. Fully addressed — the requested experiment has been run, reported with pre-registered readings, and produces a positive result.

W3 — Scope: one dataset, one autoencoder family

“The study covers only one dataset and one autoencoder family.” (R2 makes the same point: “unclear whether the same conclusions hold broadly for latent image, video, or other flow-matching models.”)

Response. We concede this. Extending to a second dataset or model family requires training new models and is out of scope for a workshop camera-ready. We prefer to state the restriction plainly rather than argue around it.

Exact change. Expand the “Scope” paragraph of §9 to state: (i) the restriction to one dataset, one autoencoder and one architecture family; (ii) that the regularity result in particular may depend on the autoencoder’s $\times 4$ temporal downsampling; (iii) that the mechanism is representation-agnostic in principle, since any lossy learned decoder between the integration space and the evaluation space should behave the same way, but that this is an expectation and not a measurement; and (iv) the concrete next test — an open-source latent flow model with a published checkpoint, measured under the same three-space protocol.

Add, in the same paragraph, the mitigating observation that the matched CDFM control isolates the latent stage *within* this setting, which is what the design was built to do and what makes the $\mathcal{Z}$ versus $\mathcal{Q}$ comparison interpretable at all.

Location. §9, “Scope” paragraph (expanded from 3 lines to approximately 8).

Status. **Remains a limitation** — stated explicitly, with the follow-up test named.

W4 — The intervals are likely too tight

“Intervals reflect only sampler noise over 3–5 replications, on a fixed clip set with a single trained model per type. They are therefore likely too tight.”

Response. Correct, and we will not defend it. Our intervals are t -based across sampling seeds with the clip set, model weights and evaluator held fixed. They therefore quantify sampling variance and *exclude* training variance and clip-set variance, both of which would widen them.

Exact change. Add a new paragraph to §9, placed immediately after “Scope”:

What the intervals cover. All reported intervals are t -based confidence intervals across sampling seeds, computed with the evaluation clip set, the model weights and the evaluator held fixed. They therefore describe sampling variance only. Training variance — which would require multiple independently trained models per type — and clip-set variance are not represented, so the intervals understate total uncertainty. This affects the precision of individual figures rather than the direction of the comparisons, all of which are paired within seed, but it should be read as a floor on uncertainty rather than an estimate of it.

Additionally, amend §4 (Experimental Setup, “Degradation measure” paragraph) to state the averaging structure explicitly: statistics are computed per clip, reduced to one value per replication, and the interval taken across replications.

Location. §9, new paragraph after “Scope”; §4, one clarifying sentence.

Status. **Remains a limitation** — conceded explicitly; no additional models were trained.

W5 — Compute checklist contradicts Appendix D.3

“The compute checklist also contradicts Appendix D.3: the checklist says a V100 and wall-clock times are reported, but the appendix says they are not.”

Response. This is a build error on our side, not a disagreement. The appendix source does contain per-experiment wall-clock figures; the submitted PDF was compiled from an out-of-sync version in which that paragraph was still a placeholder. We are grateful it was caught — the inconsistency undermines confidence in the reporting generally.

Exact change. Restore the compute paragraph in the appendix, stating that every measurement is inference-only on frozen checkpoints and runs on a single NVIDIA V100, with indicative wall-clock costs at 512 clips: replicated curvature and correlation study, five seeds, ≈ 18 minutes; step-schedule comparison, ≈ 50 minutes; reflow and guidance comparison, three seeds, ≈ 15 minutes; gain-disentangling study, ≈ 15 minutes; pair generation for the reflow student, 14 minutes, and its fine-tune ≈ 70 minutes. Note also that the project as a whole consumed considerably more compute than these figures, in preliminary and abandoned experiments not reported. Then rebuild the checklist and the appendix from the same source and verify agreement before submission.

Location. Appendix D.3 (restored paragraph); checklist item on compute (verified against it).

Status. **Fully addressed.**

2 Reviewer 2 (Rating 4, Confidence 4)

W1' — The theoretical insight is expected

“Since numerical integration is performed in the integration space, it is natural that Euler discretization error is governed by the dynamics in that space, while downstream mappings enter through sensitivity. The theoretical result mainly formalizes this intuition and is used qualitatively.”

Response. We agree, and would go further: the analysis is included precisely *because* it makes the empirical result unsurprising. An explained finding is stronger than an unexplained correlation, and we would rather present a textbook bound honestly than dress it up. The revision repositions the proposition rather than defending it.

Exact change. Beyond the rewording under R1/W1, add one sentence at the end of §3: “We include this analysis not as a novel result — it is a standard bound — but because it identifies in advance which quantities can influence the error, and thereby makes the empirical separation in §6 an expected outcome rather than a coincidence.”

Correspondingly, remove any framing in §1 that presents the analysis as a contribution. The contribution list should read: (i) the regularity result, (ii) the predictive separation, (iii) the scheduling and reflow interventions.

Location. §3 (1 new sentence); §1 contribution list (revised to exclude the bound).

Status. Fully addressed.

W2' — Experimental scope is limited

Identical in substance to R1/W3. Handled by the same revision to §9. We additionally name the concrete next test there, which responds to this reviewer’s specific framing (“latent image, video, or other flow-matching models”).

Status. Remains a limitation.

W6 — Reflow does not isolate geometry

“Reflow changes the velocity field and training coupling in addition to straightening the trajectory.”

Response. Partly answered in the submission, and we should have made that clearer; the remainder we concede.

What is already controlled: the reflow student must be sampled at guidance 1.0, because its training pairs were generated with classifier-free guidance applied. Comparing it against a teacher at guidance 2.5 would therefore confound straightening with the guidance scale. We ran the teacher at 1.0, 1.5 and 2.5 for exactly this reason, and the control resolves the confound in the direction opposite to the obvious worry: at matched guidance the improvement attributable to straightening is *larger* ($17.17 \rightarrow 0.75$ at one step) than the raw comparison against the guided teacher suggests ($7.36 \rightarrow 0.75$).

What remains confounded: reflow also alters the noise-sample coupling and retrains the velocity field. Isolating straightness alone would require an intervention that modifies trajectory curvature without retraining, which we do not have.

Exact change. In §8, immediately after the guidance-scale paragraph, add: “Two changes are bundled in this comparison. The guidance confound is controlled by the matched guidance arm above. The change in noise-sample coupling is not: reflow retrains the velocity field, so the comparison shows that an intervention which straightens the trajectory improves few-step quality, not that straightening alone is responsible. Isolating the geometric effect would require an intervention that alters curvature without retraining.”

Add a corresponding item to §9: “The reflow comparison bundles straightening with a change of training coupling; only the guidance confound is controlled.”

Location. §8 (1 new paragraph after the guidance discussion); §9 (1 limitation item).

Status. Partially addressed — the guidance confound is controlled and reported; the coupling confound is conceded.

W8 — Interventions are suggestive rather than conclusive

“The interventions support the geometric interpretation, but do not isolate the effect of geometry alone.”

Response. This is the general form of W6 and W2. The scheduling half is now substantially stronger: the curvature experiment intervenes on *nothing but* the timestep grid — the model, the weights, the noise and the evaluator are all fixed — so the A versus B comparison isolates the geometry used to build the schedule and nothing else. That is a clean isolation, and it is the one the reviewer asked for under their scheduling point.

The reflow half remains confounded, as conceded above.

Exact change. Add to the §7 discussion: “Because arms A and B differ only in the space in which the curvature profile is measured — the model, weights, noise and evaluator are identical — the A–B comparison isolates the effect of the geometry used to construct the schedule. This is a stricter isolation than the reflow experiment of §8 permits.”

Location. §7 (1 sentence in the discussion); cross-referenced from §9.

Status. **Partially addressed** — scheduling now isolates geometry; reflow does not.

W7 — MPJPE is not defined in the main text

“The paper is well written (although MPJPE is not defined in the main text).”

Response. Correct; it is defined only in the appendix.

Exact change. In §4, “Degradation measure” paragraph, expand the first sentence to define it on first use: “... measuring the mean per-joint position error (MPJPE), the mean over valid frames and joints of the Euclidean distance between corresponding joint positions.”

Location. §4, “Degradation measure”, sentence 2.

Status. **Fully addressed.**

3 Additional change arising from the new experiment

A new link between §5 and §7

Not requested by either reviewer, but produced by the experiment run for W2, and we propose to include it.

Arm C — the decoded-space curvature grid calibrated *per clip* — recovers 76% of the A–B gap on LFM at four steps (-0.743 ± 0.089 against A–B of -0.978 ± 0.068). Two controls make this interpretable: a same-noise oracle is no better than C, so the gain is prompt-level rather than noise leakage; and per-clip calibration in $\mathcal{Z}$ adds nothing over the shared $\mathcal{Z}$ grid, because $\mathcal{Z}$ is already regular.

This gives the regularity collapse of §5 a measurable practical cost, exactly where the collapse occurs and nowhere else. §5 and §7 currently stand apart in the paper with no connection between them; this supplies one.

Exact change. Add a short subsection at the end of §7, “Per-clip calibration and the cost of lost regularity”, reporting arm C and both controls, and closing with a sentence referring back to §5. Add a forward reference at the end of §5: “§7 shows that this loss of regularity has a measurable cost: a shared schedule derived from the decoded geometry is worse than one calibrated per clip, whereas in the integration space per-clip calibration adds nothing.”

Location. §7 (new final subsection, $\approx$ half a page); §5 (one forward-reference sentence).

Status. **Fully addressed** — new positive result, with controls.

4 What we will not change

Both reviewers credited the transparency of the reporting: R1 cites the flagging of the CDFM segment-midpoint gain that *failed* its robustness check, the reporting of the residual FID floor in the stratified analysis, and the refusal to over-interpret orderings of dependent correlations. These will be preserved verbatim. There is a natural temptation when revising to tidy such passages away; doing so would remove precisely what the reviewers valued.

5 Order of work

1. Restore the appendix compute paragraph and rebuild the checklist from the same source (W5).
2. Rewrite §7 around the curvature experiment; replace Table 4; add the profile and grid figures (W2, W8).

3. Add the per-clip subsection and the §5 forward reference (new result).
4. Reword Proposition 1 claims throughout, and revise the §1 contribution list (W1, W1').
5. Add the §8 confound paragraph and its §9 item (W6).
6. Expand §9 Scope; add the “What the intervals cover” paragraph (W3, W2', W4).
7. Define MPJPE in §4 (W7).

Item 2 is the substantial one. Items 1 and 7 take minutes. Everything except item 2 is writing and should be completed regardless of any further experimental work.